

RESEARCH THESIS

Predicting Next-Quarter Pharmaceutical Equity Direction

A chronological comparison of eight classification models

29 companies • Market and financial-statement predictors
Held-out observations: 2024 Q1-2026 Q1

September 2026

Completed computational study

The analysis reconstructs and corrects the supplied project, then evaluates all eight classifiers on a common chronological test panel. Results concern directional classification under the stated data and timing assumptions; they are not a trading-profitability claim.

Abstract

This study evaluates whether historical market variables and financial-statement ratios distinguish positive from non-positive next-quarter equity returns in a pharmaceutical and biopharma sample. A reproducible reconstruction of the supplied analytical project yields 3,433 company-quarter records for 29 companies: 15 Indian and 14 non-Indian. After purging 58 observations whose target horizons cross fitting or selection boundaries, the experiment uses 2,911 training observations, 203 validation observations and 261 held-out test observations. Sixteen non-redundant, training-observed predictors are retained from 22 candidate definitions. Financial data are aligned through assumed reporting lags and staleness limits; numerical imputation, percentile clipping and scaling are fitted only on training rows. Logistic Regression, Random Forest, Gaussian Naive Bayes, Gradient Boosting, SVM, MLP, KNN and Decision Tree are evaluated using saved classes and continuous scores. SVM is selected by validation balanced accuracy (53.5%), with validation ROC-AUC as a tie-breaker. On the test set it achieves 55.2% accuracy, 55.6% sensitivity, 54.5% specificity, 0.589 F1 and 0.520 ROC-AUC. The training-majority baseline reaches 57.9% accuracy by always predicting the positive class. No fitted model exceeds that accuracy. The selected SVM provides the best test balanced accuracy among the eight fitted models but only weak ranking discrimination. Extensive missing fundamentals, unequal historical coverage and unavailable point-in-time statement vintages limit interpretation. The evidence supports a cautious conclusion of limited out-of-sample directional signal in this implemented design.

Keywords: pharmaceutical equities; directional prediction; chronological validation; binary classification; financial information timing; model comparison.

Executive brief

Research problem. The task is to classify whether a company's next calendar-quarter price return is positive. The objective is comparative evaluation, including the balance between missed positive quarters and false positive forecasts, rather than a claim of market timing or investment profitability.

Completed work. The study rebuilds prices, aligns annual and quarterly financial records, defines the feature matrix, fits eight estimators, and saves every validation/test prediction. All reported metrics are recalculated from those outputs. The final dataset and scripts accompany the report.

Main result. SVM is preferred under the predeclared validation rule. Its test confusion matrix contains 60 true negatives, 50 false positives, 67 false negatives and 84 true positives. Its balanced accuracy is 55.1%, versus 50.0% for the constant baseline. Nevertheless, the baseline makes seven fewer total classification errors: 110 rather than 117. Preference therefore depends on recognising both outcome classes, not on an accuracy victory.

Interpretation. Tree ensembles tend to overpredict positive quarters; Naive Bayes errs in the opposite direction. SVM offers the most even sensitivity-specificity balance in the held-out sample. Its AUC remains close to 0.5, so neither broad superiority nor economic usefulness is established.

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1 Research problem and analytical contribution

1.1 Problem definition

Pharmaceutical equity values reflect firm-specific prospects and broad market conditions. The available data represent prices and accounting outcomes, not direct measures of regulatory announcements, patent events, trial outcomes or investor expectations. This creates a specific empirical question: can a common set of lagged numerical predictors distinguish next-quarter price direction across the supplied company universe? The observation unit is a company at a calendar-quarter end, and the prediction is evaluated against the subsequent complete quarter.

A positive outcome means a strictly positive price return. A non-positive outcome combines negative and exactly zero returns. The design is pooled across firms. Company and region identifiers are retained for joining and description but are excluded from the predictors, so the models cannot use an explicit company label to memorise unconditional firm outcomes. This does not eliminate firm-specific patterns in the remaining variables.

1.2 Research questions

The analysis addresses three questions. First, how do eight distinct classifier families perform on the same future observations? Second, does any apparent accuracy advantage survive comparison with a training-defined majority baseline and class-sensitive metrics? Third, do probabilities or oriented decision scores separate positive and non-positive outcomes, even where the native decision threshold produces uneven error rates?

The contribution is an implemented comparative experiment with an explicit data-to-prediction chain. Each model receives the same processed training matrix and the same validation/test membership. Confusion matrices, score-based ROC curves and model diagnostics make the strengths and limitations observable. This is a corrected reconstruction of the later supplied implementation, not an assertion that its outputs reproduce an earlier historical thesis table.

1.3 Relevant literature and scope

The estimator set deliberately spans simple additive decision rules, local-neighbour methods, conditional-density classification and nonlinear models. Random forests aggregate randomised trees (Breiman, 2001), while gradient boosting adds successive weak learners to improve a loss function (Friedman, 2001). Support-vector methods construct separating surfaces with margin-based regularisation (Cortes and Vapnik, 1995). These differences motivate empirical comparison; none establishes predictive success for this particular financial sample.

The implementation uses scikit-learn’s estimator interfaces (Pedregosa et al., 2011). Its guidance on training-fitted transformations informs the preprocessing isolation used here (scikit-learn developers, 2025a). ROC analysis complements threshold-dependent classification measures because it examines the ranking induced by continuous outputs (Fawcett, 2006). The report applies those tools to its own observations and does not borrow empirical conclusions from unrelated applications.

The study does not estimate causal effects, transaction-cost-adjusted returns or an investable portfolio. It does not compare newly unseen companies with previously observed companies, because companies recur across the chronological partitions. Its

external validity is consequently about later observations for this selected universe under the stated reconstruction rules.

2 Data and company-quarter construction

2.1 Sample and source interpretation

The later package defines 30 candidate pharmaceutical/biopharma companies. Its daily-price loader has usable CSV inputs for 29: 15 Indian and 14 non-Indian. Laboratorios Rovi contributes no daily CSV observations through that package path and is excluded from this implemented panel. This is an implementation-specific coverage statement, not a claim that Rovi lacks market data in general. Appendix A identifies every represented company and its training coverage. Zenotech remains represented through prices, although the package's CSV financial loader does not parse its spreadsheet-only financial material.

The reconstruction reads the package's reorganised financial and daily-market inputs. It retains the implemented special price-source rules for Biocon, Ajanta, Marksans, Strides and Sanofi. For Sanofi, the country-tagged Paris/EUR block is selected rather than the US ADR block. For the adjusted-price overrides, the specified supplement takes priority where present. Otherwise adjusted close is preferred when the selected source block contains it. Source definitions and selection decisions are saved with the run.

Dates are interpreted as their recorded local calendar dates. Invalid, missing or non-positive prices are removed before endpoint selection. Within the selected price basis, overlapping dates are resolved deterministically by the source with the longest valid coverage, then source name. Volume is retained rather than discarded. These choices avoid conflating a parsing omission with a genuinely unavailable variable. They do not guarantee that every vendor adjustment or predecessor-company history is economically comparable.

2.2 Quarter endpoints and target

For each company, calendar-quarter ends are generated only within the observed daily coverage. The last trading observation on or before the current quarter end supplies $P(t)$; the corresponding observation for the next complete quarter supplies $P(t+1)$. Both endpoints must be no more than ten calendar days before their quarter end. A row without a following complete quarter is not labelled. The target uses the price basis selected for that company, without a market-return adjustment.

$$r(t+1) = P(t+1) / P(t) - 1; \quad y(t) = 1 \text{ if } r(t+1) > 0, \text{ otherwise } 0. \quad (1)$$

The target is therefore a next-quarter price-direction label, not an excess-return or risk-adjusted-return label. For adjusted-close sources it incorporates that source's adjustment convention. No assumption is made that the result is a tradable total return. Exactly zero returns remain in the non-positive class. Endpoint prices, endpoint dates and forward returns are saved alongside every panel row, permitting direct recalculation.

2.3 Coverage and chronological partitions

Table 1. Chronological estimation and evaluation sample. Positive means next-quarter return > 0.

Partition	Observations	Positive	Non-positive	Observation dates
train	2911	1726	1185	1962-03 to 2021-09
validation	203	117	86	2022-03 to 2023-09
test	261	151	110	2024-03 to 2026-03

Table 1 shows the common fitting and evaluation partitions. The training period begins in 1962 for the longest available company history and expands as additional firms enter the data. This is an unbalanced historical panel. The validation and test panels contain 29 companies in each included quarter. The most recent labelled observation is 2026 Q1, whose outcome uses the endpoint of 2026 Q2.

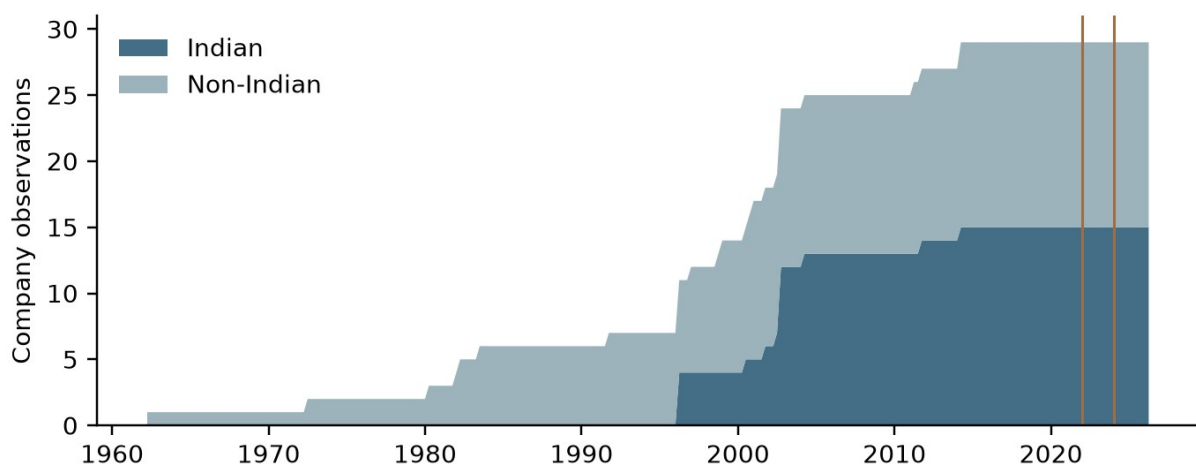


Figure 1. Company observations over time in the retained partitions. Vertical markers denote the 2022 and 2024 observation-date boundaries. Unequal early coverage is part of the actual sample.

Figure 1 makes the changing sample density explicit. Long-lived series contribute many more training observations than later listings. The use of all available history follows the later package, rather than imposing the older draft’s claimed balanced 2014–2026 sample. Because early records rarely have corresponding fundamentals, this choice also changes the missingness distribution; its implications are examined in Section 5.

A chronological observation split alone does not isolate future-return labels. The 2021 Q4 observation would use a 2022 Q1 outcome, and the 2023 Q4 observation would use a 2024 Q1 outcome. Both boundary quarters are purged for all 29 companies, removing 58 records. Training ends at 2021 Q3 and validation ends at 2023 Q3; their labels are known before the next partition begins. Models are not refitted on validation data after selection.

3 Methodology and model specifications

3.1 Financial information alignment

The financial parser accommodates Capital IQ-style period columns, date-header statement supplements and derived long-format extracts. The source-field aliases supplied by the package map revenues, income, assets, equity and current balance-sheet quantities into common fields. Monetary amounts are converted from reported

thousands, millions or billions into base-currency units before cross-statement ratios or growth rates are calculated. Unrecognised unit scales are not assigned invented conversions.

For ordinary quarterly records, availability is the financial period end plus 61 days; annual records use plus 121 days. These preserve the package’s strict “after 60/120 days” convention. The latest eligible financial-period row is selected only if its period is at most 200 days old for quarterly data or 550 days old for annual data. A missing field in that selected row remains missing, rather than being silently backfilled from arbitrarily old periods. No real-time release database is available to verify each assumed publication lag.

Revenue growth uses the nearest same-frequency period approximately one year earlier, within 45 days of that anniversary. Profit margins use income divided by revenue; balance-sheet ratios use aligned balances. Return-on-assets, return-on-equity and asset turnover are period-flow-to-ending-stock proxies, not trailing-twelve-month or annualised ratios. Annual and quarterly fallback can therefore produce different horizons. These features must be read as implemented accounting proxies, with the frequency limitation retained in the interpretation.

3.2 Final predictor set

The supplied code contains 22 concrete candidate definitions. Three are entirely missing in the training data: debt-to-equity, price-to-book and the price-to-earnings proxy. They are removed because a training median does not exist. Three other columns are exact duplicates of earlier definitions: 3-month momentum duplicates the 91-day return, 6-month momentum duplicates the 182-day return, and the “quick ratio proxy” duplicates current ratio. Their removal is determined from training data and avoids counting duplicated information as distinct predictors.

The retained set contains nine market features and seven financial features. Return horizons and moving-average/volatility windows are measured in calendar days, not in a fixed number of trading sessions. Volatility is the sample standard deviation of daily percentage changes within the calendar window and is not annualised. All feature-price calculations receive only observations dated on or before the observation quarter. The original code’s extra-day access is therefore disabled by construction. Appendix B gives the complete candidate dictionary and retained/excluded status.

Table 2. Retained feature families and implemented information.

Family	Count	Variables
Market	9	91-, 182- and 365-day returns; 20-, 50- and 200-day price-to-mean deviations; 60- and 120-day volatility; 60-day volume change
Financial	7	Revenue growth; net and operating margins; return on equity; return on assets; current ratio; asset turnover

3.3 Training-only preprocessing

Infinite or undefined numerical values are treated as missing. Active columns are determined exclusively from training observations, and missing values in every partition are filled with the corresponding training median. The 1st and 99th percentiles of observed training values define clipping bounds. After imputation and clipping, a StandardScaler is fitted to training rows and applied unchanged to

validation and test rows. This controls extreme values and places distance-sensitive and gradient-based models on comparable numerical scales.

The same transformed 16-column matrix is supplied to all eight estimators. A monotone affine scaling does not supply new information to the trees; its use across estimators simplifies the shared processing contract. Model objects retain feature order, fitted medians, clipping bounds and scaling parameters. These transformations are reconstructed independently from the saved training panel in the verification checks. No oversampling or synthetic observations are used.

3.4 Estimator settings and class weights

Table 3. Fixed estimator specifications. Complete resolved parameters are supplied in Appendix C and the run files.

Model	Core specification	Class treatment
Logistic Regression	L2; C=1; lbfgs; max_iter=1,000	Balanced
Random Forest	100 trees; Gini; sqrt features; no depth cap	Balanced
Naïve Bayes	GaussianNB; var_smoothing=1e−9	Empirical priors
Gradient Boosting	100 stages; learning_rate=0.1; depth=3; log loss	Unweighted
SVM	RBF; C=1; gamma=scale; probability=True	Balanced
Neural Network / MLP	100 ReLU units; Adam; alpha=0.0001; max_iter=2,000	Unweighted
KNN	5 neighbours; uniform voting; Minkowski p=2	Unweighted
Decision Tree	Gini; no depth cap; min_samples_leaf=1	Balanced

Table 3 records the implemented settings. They are inherited from the supplied later pipeline, not selected by a new grid search. Random seeds are 42 where the estimator exposes a seed. Balanced weights are used only where configured by that pipeline; this asymmetry is disclosed rather than represented as a controlled study of algorithm architecture alone. The MLP uses its default Adam stopping criterion without internal early stopping; the completed fit stopped after 779 iterations. Logistic Regression required 12 iterations. No training warnings were recorded.

3.5 Validation selection and protected test evaluation

The primary selection criterion is validation balanced accuracy, the average of positive-class sensitivity and negative-class specificity. If tied, validation ROC-AUC breaks the tie, followed by model name for deterministic ordering. This rule is recorded before the corrected test evaluation and responds to the requirement to assess both classes. It does not presume a monetary cost ratio for missed positive quarters versus false positive forecasts. No decision threshold is tuned on the test set.

All eight estimators retain their native predict decision rule. For SVM, the decision-function sign determines the class; its correctly oriented decision score is used for ROC-AUC. The internally estimated SVM probabilities are also saved, but need not imply the same 0.5 threshold as the native class decision. For the other models, positive-class predict_proba values supply the ROC score. Learned class ordering is checked to be [0, 1]. All test predictions and scores are then saved row by row.

The selected model is not the estimator with the largest observed test accuracy. Selection uses the separate validation partition, and the test table reports all eight frozen models as a comparative evaluation. The archive already exposed outcomes for

related data, so this exercise cannot be described as a fully blinded prospective experiment. The corrected protocol isolates fitting and selection in code, while acknowledging that prior project results were available to the reconstruction process.

3.6 Evaluation definitions

Table 4. Binary evaluation definitions. Up is the positive class; undefined precision/F1 is reported as zero by the stated implementation.

Measure	Definition	Interpretation
Accuracy	$(TP + TN) / N$	Overall fraction correct
Precision	$TP / (TP + FP)$	Fraction of positive forecasts that are correct
Recall / sensitivity	$TP / (TP + FN)$	Fraction of positive outcomes detected
Specificity	$TN / (TN + FP)$	Fraction of non-positive outcomes detected
F1	$2TP / (2TP + FP + FN)$	Positive-class precision-recall balance
Balanced accuracy	$(\text{Sensitivity} + \text{Specificity}) / 2$	Equal weighting of the two outcome classes
ROC-AUC	$P(\text{score}(\text{up}) > \text{score}(\text{non-up}))$, with half credit for ties	Ranking discrimination across thresholds

The confusion matrix is ordered with actual classes in rows and predicted classes in columns: $[[TN, FP], [FN, TP]]$. A false positive is a forecast of an increase followed by a non-positive return. A false negative is a forecast of a non-positive outcome followed by an increase. Neither error directly measures money lost because return magnitudes, execution, exposure and trading costs are not modelled. Table 4 separates the statistical meanings rather than treating all metrics as interchangeable.

ROC curves use actual continuous outputs and are not inferred from a scalar AUC or reconstructed from class predictions. For the unrestricted Decision Tree, terminal-leaf outputs may be effectively discrete, so the resulting curve has few operating points. The AUC remains a valid ranking calculation for those saved scores, but it is not evidence of finely resolved probability calibration (Fawcett, 2006).

4 Empirical results

4.1 Validation evidence and selected model

Table 5. Validation performance on 203 observations; settings are fixed, not grid-searched.

Model	Accuracy	Precision	Recall	Specificity	F1	Bal. acc.	AUC
Logistic Regression	0.488	0.563	0.496	0.477	0.527	0.486	0.541
Random Forest	0.542	0.582	0.726	0.291	0.646	0.509	0.487
Naive Bayes	0.478	0.604	0.274	0.756	0.376	0.515	0.478
Gradient Boosting	0.493	0.550	0.658	0.267	0.599	0.463	0.492
SVM	0.532	0.612	0.513	0.558	0.558	0.535	0.578
Neural Network / MLP	0.527	0.590	0.590	0.442	0.590	0.516	0.531
KNN	0.502	0.566	0.590	0.384	0.577	0.487	0.481
Decision Tree	0.507	0.568	0.607	0.372	0.587	0.489	0.489

Model	Accuracy	Precision	Recall	Specificity	F1	Bal. acc.	AUC
Majority baseline	0.576	0.576	1.000	0.000	0.731	0.500	0.500

SVM has the highest validation balanced accuracy, 0.535, and the highest validation AUC, 0.578, among the eight fitted classifiers. Its validation sensitivity is 0.513 and specificity is 0.558. Random Forest has slightly higher ordinary validation accuracy, 0.542 versus 0.532, but its specificity is only 0.291. Selecting Random Forest solely by accuracy would therefore favour a much stronger positive-class bias. Under the declared rule, SVM is preferred before the test comparison.

The constant positive baseline has 0.576 validation accuracy and 0.731 F1, both higher than any fitted classifier. Yet its specificity is zero. This demonstrates why accuracy and positive-class F1 alone are inadequate for the present two-class objective. SVM's validation advantage in balanced accuracy is modest; the selection rule identifies a preferred candidate within this experiment, not a definitively superior forecasting technology.

4.2 Common held-out test comparison

Table 6. Held-out test performance on the same 261 observations. Values are proportions; all AUCs use saved continuous scores.

Model	Accuracy	Precision	Recall	Specificity	F1	Bal. acc.	AUC
Logistic Regression	0.494	0.573	0.497	0.491	0.532	0.494	0.514
Random Forest	0.533	0.581	0.689	0.318	0.630	0.503	0.477
Naive Bayes	0.425	0.510	0.166	0.782	0.250	0.474	0.503
Gradient Boosting	0.510	0.561	0.695	0.255	0.621	0.475	0.491
SVM	0.552	0.627	0.556	0.545	0.589	0.551	0.520
Neural Network / MLP	0.487	0.555	0.570	0.373	0.562	0.471	0.466
KNN	0.467	0.537	0.576	0.318	0.556	0.447	0.460
Decision Tree	0.487	0.553	0.583	0.355	0.568	0.469	0.469
Majority baseline	0.579	0.579	1.000	0.000	0.733	0.500	0.500

The test set contains 151 positive and 110 non-positive outcomes. The training majority is positive (1,726 of 2,911 training observations), so the baseline predicts “up” for every test row without learning the test majority. It obtains 151 correct predictions out of 261. SVM obtains 144; Random Forest obtains 139. No fitted classifier exceeds the baseline's 0.579 accuracy. The strongest fitted accuracy and balanced accuracy both belong to the validation-selected SVM, but this agreement is an observed test result, not the selection procedure.

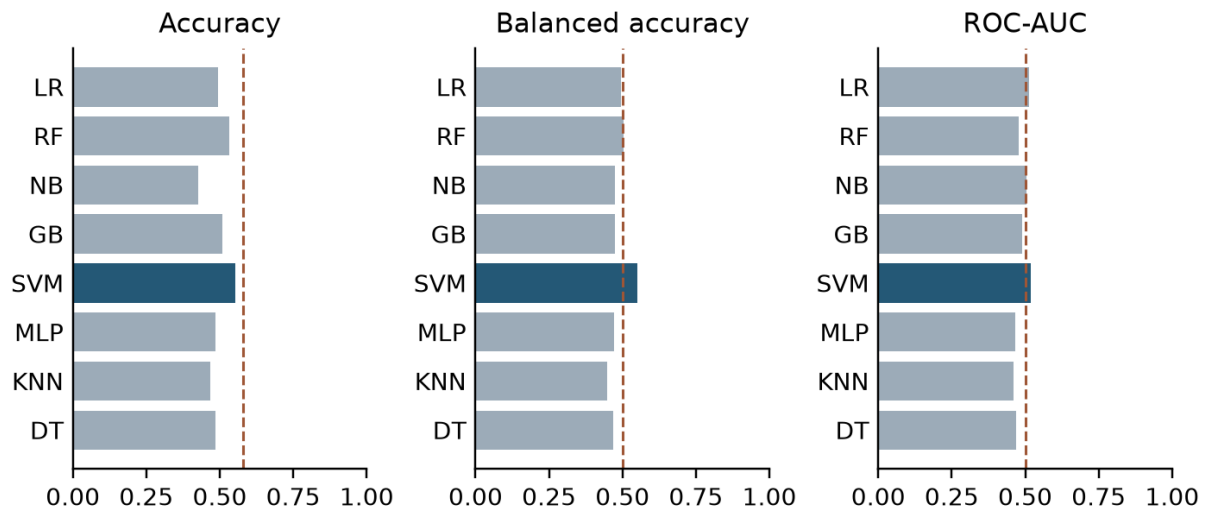


Figure 2. Held-out accuracy, balanced accuracy and AUC for all eight models. Dashed lines show the training-majority baseline. SVM is highlighted because it was selected on validation data. LR=Logistic Regression; RF=Random Forest; NB=Naive Bayes; GB=Gradient Boosting; DT=Decision Tree.

Figure 2 shows that SVM's improvement over a 0.5 balanced-accuracy reference is approximately five percentage points, whereas its AUC is only about two points above 0.5. Thus the saved operating threshold produces a more favourable balance than the broad ranking measure might suggest. All remaining fitted AUCs are between 0.460 and 0.514. No confidence interval or formal superiority test is claimed from these point estimates.

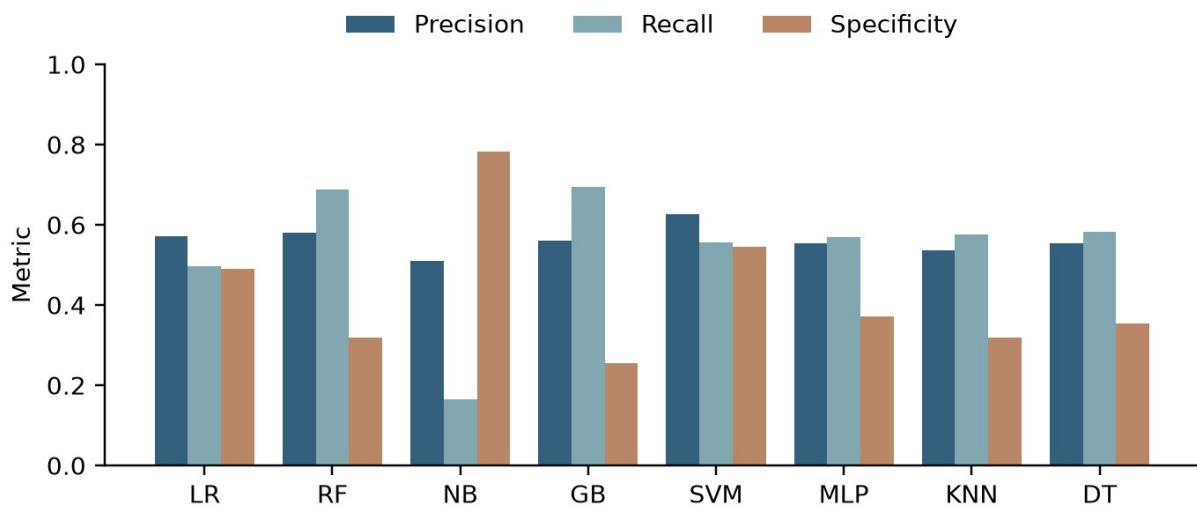


Figure 3. Precision, sensitivity and specificity at each model's saved decision rule. The imbalance between class-specific detection rates is visible even when overall accuracy differs little.

Figure 3 highlights two opposing error patterns. Naive Bayes detects many non-positive quarters but misses most positive ones. Gradient Boosting and Random Forest detect more positive quarters while accepting many false positives. SVM has the smallest absolute gap between sensitivity and specificity among the stronger-performing candidates. Those differences are central to interpretation, because the same accuracy can conceal very different decisions.

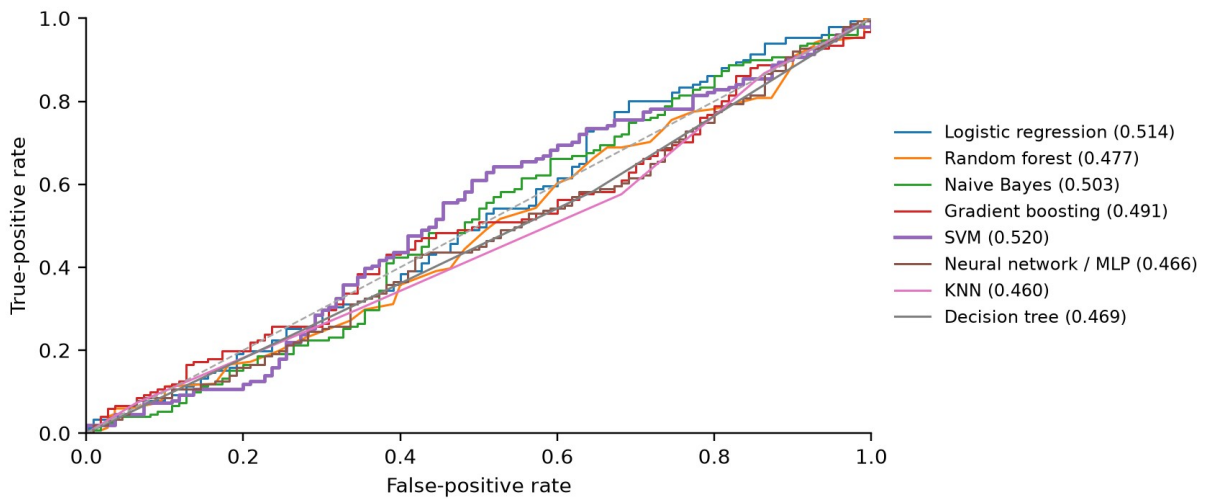


Figure 4. Test ROC curves and AUCs from positive-class probabilities, except SVM, which uses its positive-oriented decision function. The diagonal is the no-ranking-skill reference.

The curves in Figure 4 lie close to the diagonal over much of their range. There is no model with consistently strong separation across thresholds. The AUCs are recalculated independently using all positive-negative score pairs with half credit for ties, and agree with the primary metric calculation. An AUC below 0.5 is reported as observed; score orientation is not flipped after seeing the test outcomes to manufacture an improved result.

4.3 Logistic Regression

Logistic Regression is evaluated on all 261 held-out rows. It produces $TN=54$, $FP=56$, $FN=76$ and $TP=75$. Accuracy is 49.4%, precision 57.3%, recall/sensitivity 49.7%, specificity 49.1%, F1 0.532 and ROC-AUC 0.514. Its balanced accuracy is 49.4%. These quantities are derived from the saved test predictions rather than transcribed from historical project prose.

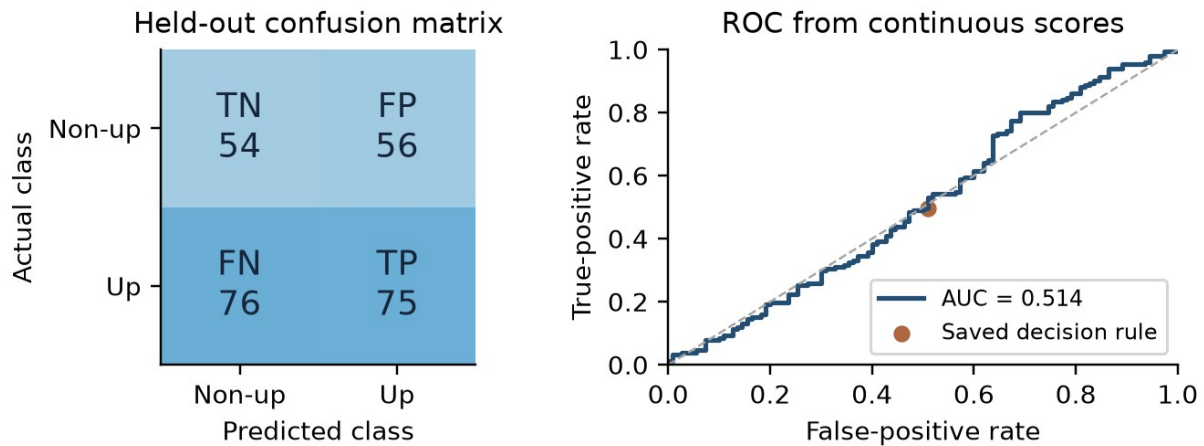


Figure 5. Logistic Regression: test confusion matrix and score-based ROC curve. Counts sum to 261; actual non-positive and positive row totals are 110 and 151. The orange point is the saved native decision rule.

The linear model produces almost equal numbers of positive and non-positive forecasts. Its sensitivity and specificity are both near one half, so the result is approximately chance-like at its native threshold. Precision exceeds 0.5 partly because positive outcomes are more common in the test sample; it remains below the majority baseline's positive precision. The model's AUC is only slightly above 0.5, indicating weak ranking information rather than a clearly useful separation.

L2 regularisation and scaling yield a numerically stable additive benchmark. The fitted coefficient signs, discussed in Section 5, describe conditional associations within the processed training matrix. They cannot establish that a variable causes subsequent returns. Its relatively even errors do not compensate for its low detection rates: balance without discrimination is not forecasting success.

Figure 5 connects the threshold-dependent errors to the continuous-score evaluation. The model's false-positive rate is 50.9% and false-negative rate is 50.3%. Its positive forecast rate is 50.2%, compared with the observed positive fraction of 57.9%. No causal or economic-loss interpretation is attached to these counts without a separate return-and-cost analysis.

4.4 Random Forest

Random Forest is evaluated on all 261 held-out rows. It produces $TN=35$, $FP=75$, $FN=47$ and $TP=104$. Accuracy is 53.3%, precision 58.1%, recall/sensitivity 68.9%, specificity 31.8%, F1 0.630 and ROC-AUC 0.477. Its balanced accuracy is 50.3%. These quantities are derived from the saved test predictions rather than transcribed from historical project prose.

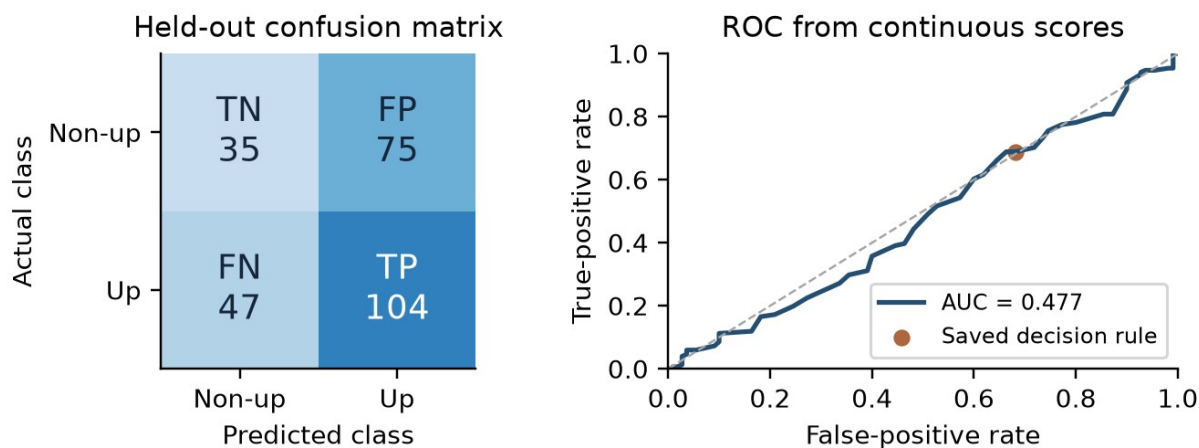


Figure 6. Random Forest: test confusion matrix and score-based ROC curve. Counts sum to 261; actual non-positive and positive row totals are 110 and 151. The orange point is the saved native decision rule.

The forest predicts positive outcomes for more than two thirds of the test rows. It detects substantially more actual positive quarters than Logistic Regression, but that gain is accompanied by 75 false positives and only 35 true negatives. Its precision is close to the constant baseline's precision, so the positive forecasts provide little enrichment over predicting an increase everywhere.

Randomised tree aggregation gives the second-highest fitted-model accuracy in this run, but balanced accuracy is almost exactly 0.5 and AUC is below 0.5. The observed threshold performance must therefore not be translated into strong ranking capability. Balanced class weights do not guarantee balanced predictions after fitting, especially when the training and test feature distributions differ.

Figure 6 connects the threshold-dependent errors to the continuous-score evaluation. The model's false-positive rate is 68.2% and false-negative rate is 31.1%. Its positive forecast rate is 68.6%, compared with the observed positive fraction of 57.9%. No causal or economic-loss interpretation is attached to these counts without a separate return-and-cost analysis.

4.5 Naive Bayes

Naive Bayes is evaluated on all 261 held-out rows. It produces $TN=86$, $FP=24$, $FN=126$ and $TP=25$. Accuracy is 42.5%, precision 51.0%, recall/sensitivity 16.6%, specificity 78.2%, F1 0.250 and ROC-AUC 0.503. Its balanced accuracy is 47.4%. These quantities are derived from the saved test predictions rather than transcribed from historical project prose.

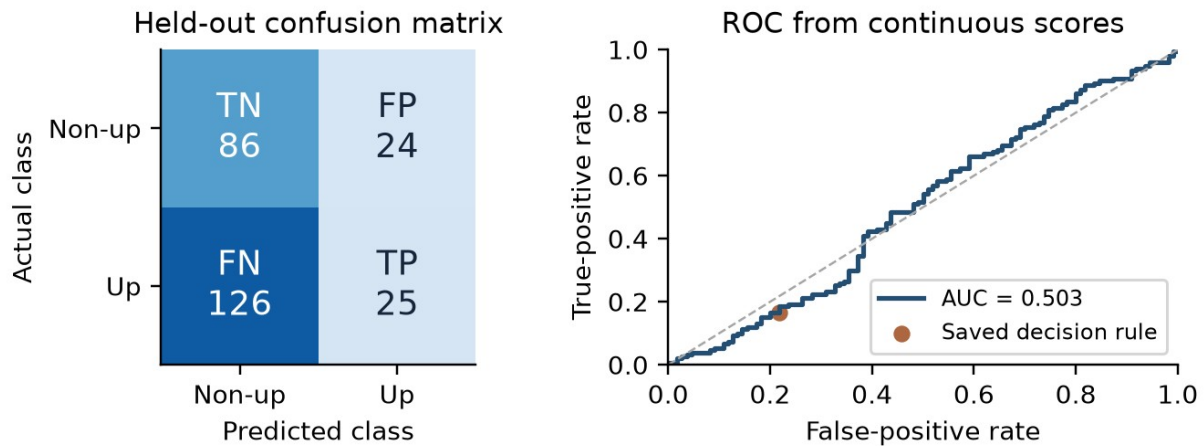


Figure 7. Naive Bayes: test confusion matrix and score-based ROC curve. Counts sum to 261; actual non-positive and positive row totals are 110 and 151. The orange point is the saved native decision rule.

Naive Bayes exhibits pronounced negative-class bias. It predicts an increase only 49 times and misses 126 of the 151 positive outcomes. Its high specificity is therefore achieved by declining to predict the positive class for most rows. The low F1 reflects the resulting poor sensitivity, rather than a numerical failure or an absent metric.

The Gaussian conditional-independence model is a deliberately simple density benchmark. Its AUC is approximately 0.5, so shifting the threshold cannot be presumed to recover a strong ordering from these scores. Even after removal of exact duplicate definitions, several market windows and financial ratios remain correlated. That is a plausible constraint on the model's assumptions, not a demonstrated causal explanation of its errors.

Figure 7 connects the threshold-dependent errors to the continuous-score evaluation. The model's false-positive rate is 21.8% and false-negative rate is 83.4%. Its positive forecast rate is 18.8%, compared with the observed positive fraction of 57.9%. No causal or economic-loss interpretation is attached to these counts without a separate return-and-cost analysis.

4.6 Gradient Boosting

Gradient Boosting is evaluated on all 261 held-out rows. It produces $TN=28$, $FP=82$, $FN=46$ and $TP=105$. Accuracy is 51.0%, precision 56.1%, recall/sensitivity 69.5%, specificity 25.5%, F1 0.621 and ROC-AUC 0.491. Its balanced accuracy is 47.5%. These quantities are derived from the saved test predictions rather than transcribed from historical project prose.

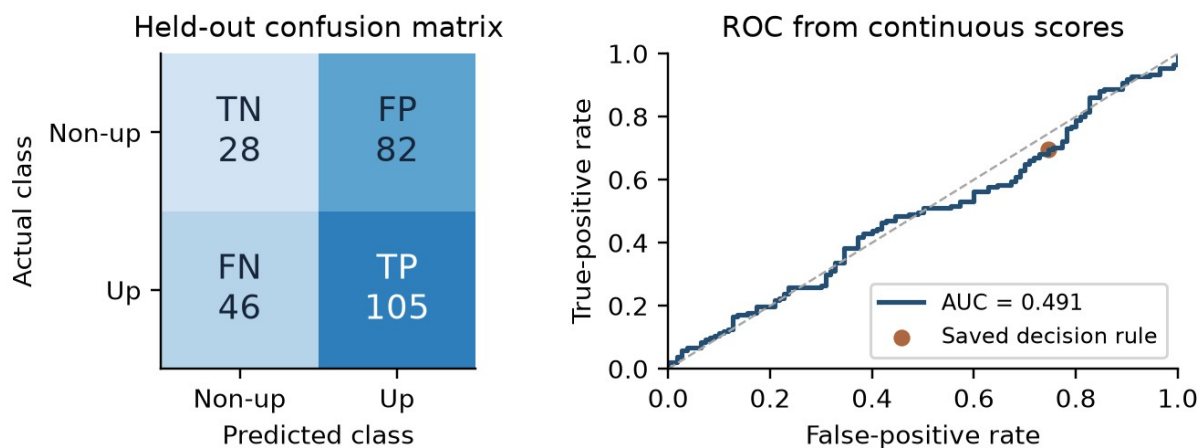


Figure 8. Gradient Boosting: test confusion matrix and score-based ROC curve. Counts sum to 261; actual non-positive and positive row totals are 110 and 151. The orange point is the saved native decision rule.

Gradient Boosting predicts “up” for nearly 72% of the test observations. Its 105 true positives give high sensitivity relative to most models, but 82 false positives reduce specificity to roughly one quarter. F1 remains comparatively high because it ignores true negatives and rewards recovery of the majority positive class.

The model adds shallow trees sequentially under log loss. In this application its nonlinear flexibility does not yield a favourable overall error balance: balanced accuracy and AUC are both below 0.5. The contrast between its positive-class F1 and its weak specificity is a concrete example of why a single summary measure can overstate usefulness. No larger ensemble or alternative learning rate was tried after observing this test result.

Figure 8 connects the threshold-dependent errors to the continuous-score evaluation. The model’s false-positive rate is 74.5% and false-negative rate is 30.5%. Its positive forecast rate is 71.6%, compared with the observed positive fraction of 57.9%. No causal or economic-loss interpretation is attached to these counts without a separate return-and-cost analysis.

4.7 SVM

SVM is evaluated on all 261 held-out rows. It produces $TN=60$, $FP=50$, $FN=67$ and $TP=84$. Accuracy is 55.2%, precision 62.7%, recall/sensitivity 55.6%, specificity 54.5%, F1 0.589 and ROC-AUC 0.520. Its balanced accuracy is 55.1%. These quantities are derived from the saved test predictions rather than transcribed from historical project prose.

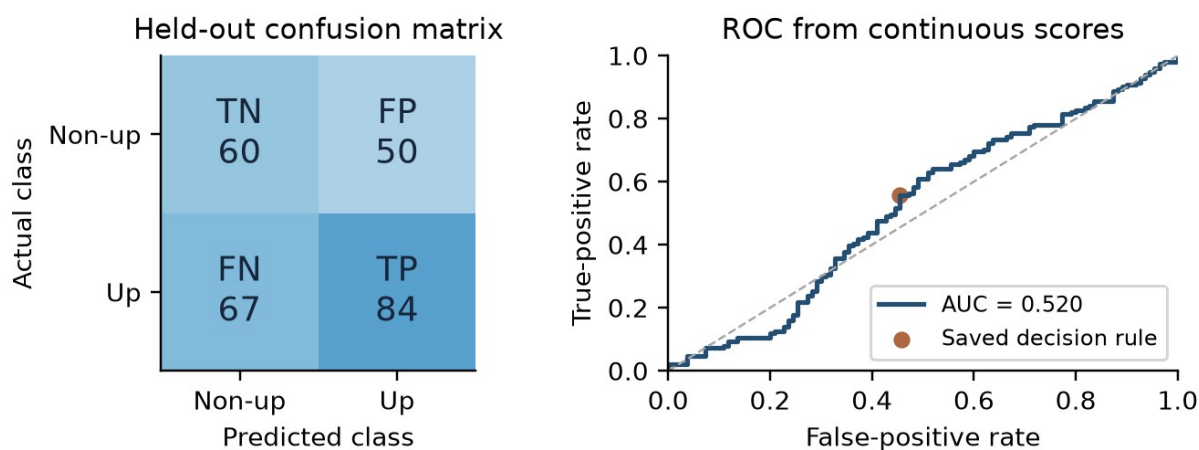


Figure 9. SVM: test confusion matrix and score-based ROC curve. Counts sum to 261; actual non-positive and positive row totals are 110 and 151. The orange point is the saved native decision rule.

SVM's native decision rule has the most even useful detection rates in this comparison: it identifies 84 of 151 positive outcomes and 60 of 110 non-positive outcomes. Its 50 false positives are considerably fewer than the tree ensembles' counts, while its 67 false negatives remain substantial. The model does not simply reproduce the positive majority and achieves the largest balanced accuracy of the fitted candidates.

The preference for SVM is justified by validation balanced accuracy and validation AUC, not by retroactively selecting the test winner. On the test data it still makes seven more total errors than the constant baseline, and its 0.520 AUC indicates weak ranking. The result supports using SVM as the preferred research comparator under equal class emphasis, not as a proven trading system. Its saved probability estimates are not asserted to be calibrated; ROC uses the oriented margin score.

Figure 9 connects the threshold-dependent errors to the continuous-score evaluation. The model's false-positive rate is 45.5% and false-negative rate is 44.4%. Its positive forecast rate is 51.3%, compared with the observed positive fraction of 57.9%. No causal or economic-loss interpretation is attached to these counts without a separate return-and-cost analysis.

4.8 Neural Network / MLP

Neural Network / MLP is evaluated on all 261 held-out rows. It produces $TN=41$, $FP=69$, $FN=65$ and $TP=86$. Accuracy is 48.7%, precision 55.5%, recall/sensitivity 57.0%, specificity 37.3%, F1 0.562 and ROC-AUC 0.466. Its balanced accuracy is 47.1%. These quantities are derived from the saved test predictions rather than transcribed from historical project prose.

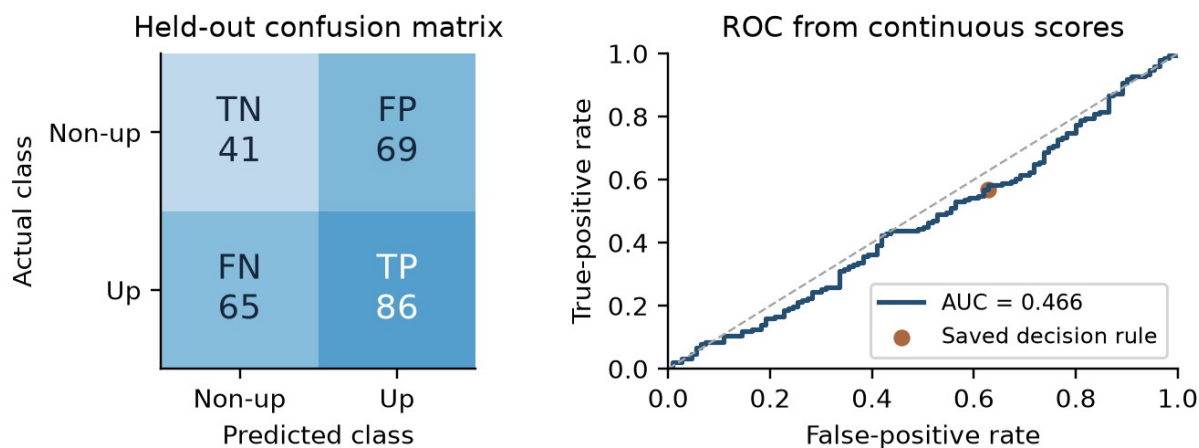


Figure 10. Neural Network / MLP: test confusion matrix and score-based ROC curve. Counts sum to 261; actual non-positive and positive row totals are 110 and 151. The orange point is the saved native decision rule.

The MLP predicts positive outcomes more often than the empirical positive fraction, but its 86 true positives do not offset 69 false positives and 65 missed increases. Its recall exceeds specificity, producing a positive-class bias despite the nonlinear architecture. Its accuracy is below one half and its balanced accuracy and AUC are also below 0.5.

The completed Adam fit stopped after 779 iterations without a recorded convergence warning. Thus the poor held-out result cannot be described merely as a failed or unfinished optimiser run. A single 100-unit architecture was retained from the supplied implementation. The evidence concerns that fixed specification and seed; it does not establish that every neural architecture would fail on every pharmaceutical dataset.

Figure 10 connects the threshold-dependent errors to the continuous-score evaluation. The model's false-positive rate is 62.7% and false-negative rate is 43.0%. Its positive forecast rate is 59.4%, compared with the observed positive fraction of 57.9%. No causal or economic-loss interpretation is attached to these counts without a separate return-and-cost analysis.

4.9 KNN

KNN is evaluated on all 261 held-out rows. It produces $TN=35$, $FP=75$, $FN=64$ and $TP=87$. Accuracy is 46.7%, precision 53.7%, recall/sensitivity 57.6%, specificity 31.8%, F1 0.556 and ROC-AUC 0.460. Its balanced accuracy is 44.7%. These quantities are derived from the saved test predictions rather than transcribed from historical project prose.

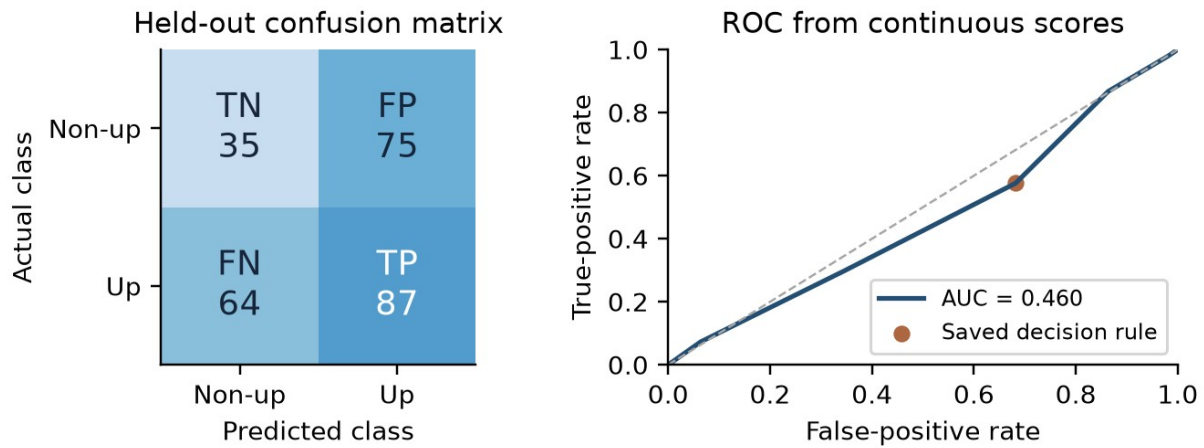


Figure 11. KNN: test confusion matrix and score-based ROC curve. Counts sum to 261; actual non-positive and positive row totals are 110 and 151. The orange point is the saved native decision rule.

KNN issues 162 positive forecasts. It catches 87 positive outcomes but incorrectly flags 75 non-positive outcomes as positive. The resulting specificity is low, and its 64 false negatives show that positive-class bias has not produced particularly high sensitivity either. Its balanced accuracy is the lowest of the eight classifiers in this run.

Distances are computed after training-only imputation, clipping and standardisation. This prevents raw numerical units from dominating the neighbour calculation, but it cannot make imputed, correlated or historically heterogeneous features locally informative. Its AUC of 0.460 comes from the actual five-neighbour vote proportions, which have only a small number of possible score levels. No post-test neighbour-count search is performed.

Figure 11 connects the threshold-dependent errors to the continuous-score evaluation. The model's false-positive rate is 68.2% and false-negative rate is 42.4%. Its positive forecast rate is 62.1%, compared with the observed positive fraction of 57.9%. No causal or economic-loss interpretation is attached to these counts without a separate return-and-cost analysis.

4.10 Decision Tree

Decision Tree is evaluated on all 261 held-out rows. It produces $TN=39$, $FP=71$, $FN=63$ and $TP=88$. Accuracy is 48.7%, precision 55.3%, recall/sensitivity 58.3%, specificity 35.5%, F1 0.568 and ROC-AUC 0.469. Its balanced accuracy is 46.9%. These quantities are derived from the saved test predictions rather than transcribed from historical project prose.

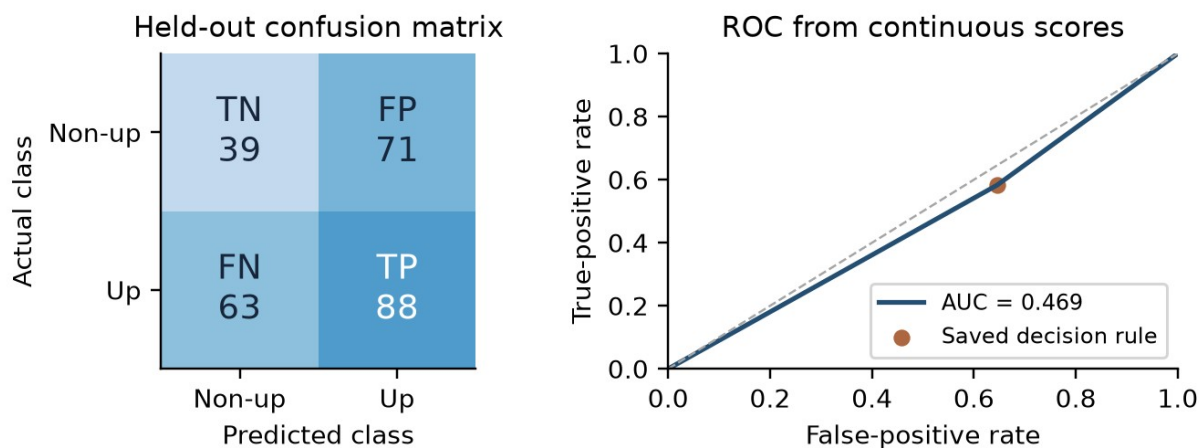


Figure 12. Decision Tree: test confusion matrix and score-based ROC curve. Counts sum to 261; actual non-positive and positive row totals are 110 and 151. The orange point is the saved native decision rule.

The unrestricted tree predicts an increase for 159 observations. It identifies 88 positive quarters but produces 71 false positives and misses 63 positive quarters. Sensitivity is higher than specificity, and the overall balance is weaker than the constant baseline's 0.5 balanced accuracy. The positive-class F1 should be read alongside those negative-class failures.

An unrestricted single tree can fit detailed partitions of the training matrix, yet its held-out score distribution provides little useful ranking. Here the AUC equals balanced accuracy because the saved terminal-leaf scores are effectively binary; this is a property of these outputs, not a general identity for all trees. The model offers a transparent structural benchmark, but no pruning parameter was optimised after seeing test performance.

Figure 12 connects the threshold-dependent errors to the continuous-score evaluation. The model's false-positive rate is 64.5% and false-negative rate is 41.7%. Its positive forecast rate is 60.9%, compared with the observed positive fraction of 57.9%. No causal or economic-loss interpretation is attached to these counts without a separate return-and-cost analysis.

5 Feature diagnostics and computational checks

5.1 Missingness and historical composition

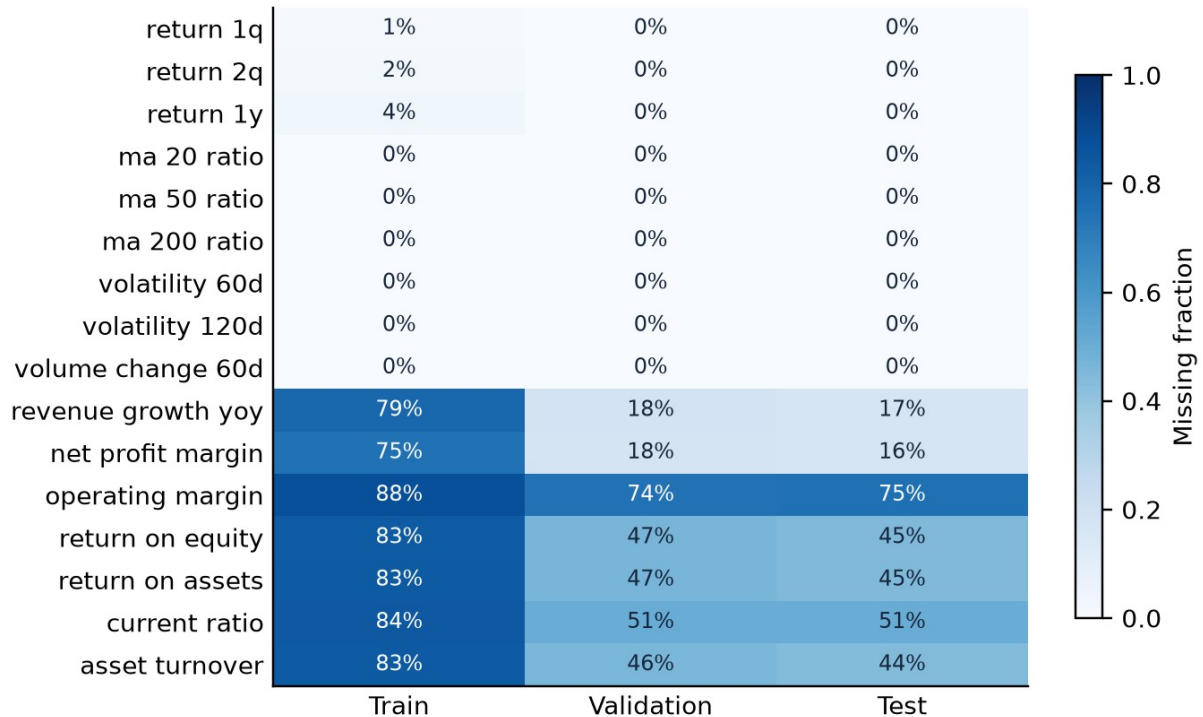


Figure 13. Observed missingness before imputation for the 16 retained predictors. Fractions are measured separately in each partition; the imputation statistics themselves use training data only.

Figure 13 shows a substantial mismatch between the historical training panel and the modern evaluation periods. Approximately 78.9% of training revenue-growth values and 75.0% of net-margin values are missing, versus 16.9% and 16.5% in the test data. Operating margin is missing for 87.6% of training rows and 75.1% of test rows. The models therefore learn with many median-filled accounting values, particularly in early history.

This is not evidence that accounting variables have no predictive relevance. It indicates that the present reconstructed matrix provides limited observations of them. Train-only imputation protects the split boundary but does not make missing values observed, remove selection effects or restore historical reporting coverage. A study restricted to a recent period or supplied with point-in-time fundamentals could behave differently; no such alternative is substituted into this final comparison.

5.2 Selected-model and coefficient diagnostics

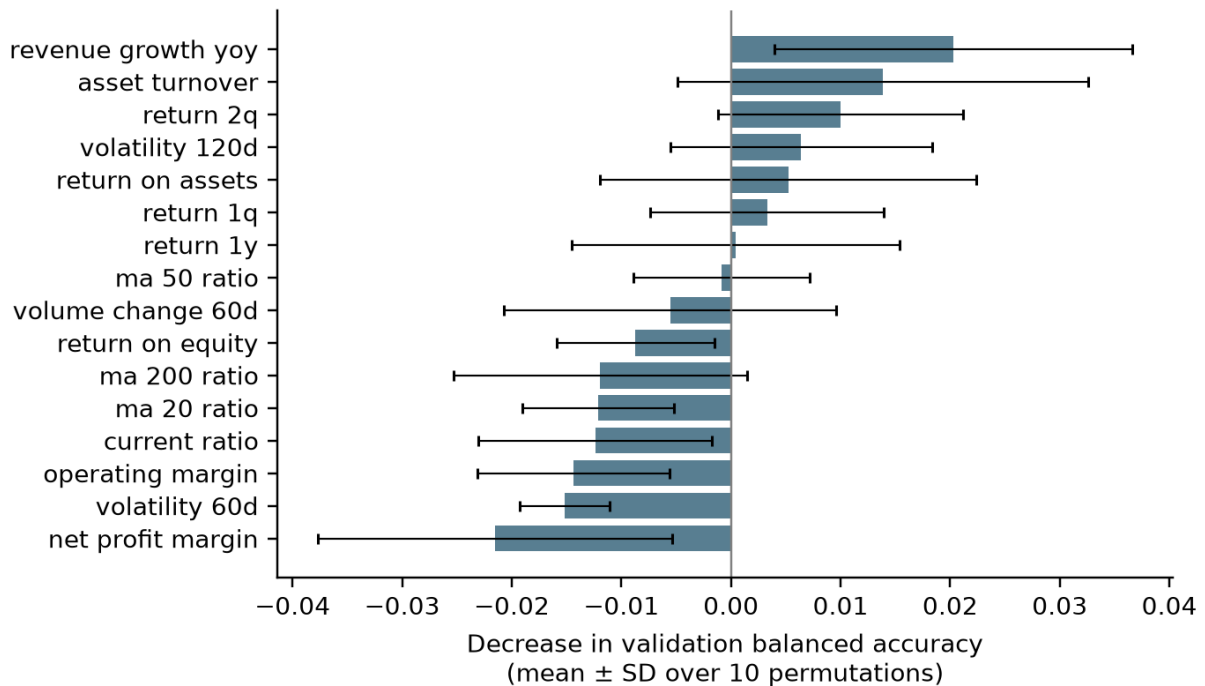


Figure 14. SVM permutation importance on validation rows, measured as the decrease in balanced accuracy over ten shuffles per feature. Error bars are shuffle standard deviations, not confidence intervals.

In Figure 14, revenue growth has the largest mean validation permutation decrease, about 0.020, followed by asset turnover at 0.014 and the 182-day return at 0.010. Several features have negative mean decreases: shuffling them improves the score in this small validation sample. The spreads are often as large as the means, so a strong or stable ranking is not established. Correlated features and median imputation further complicate attribution. This diagnostic was not used to revise the final feature set or retune the selected model.

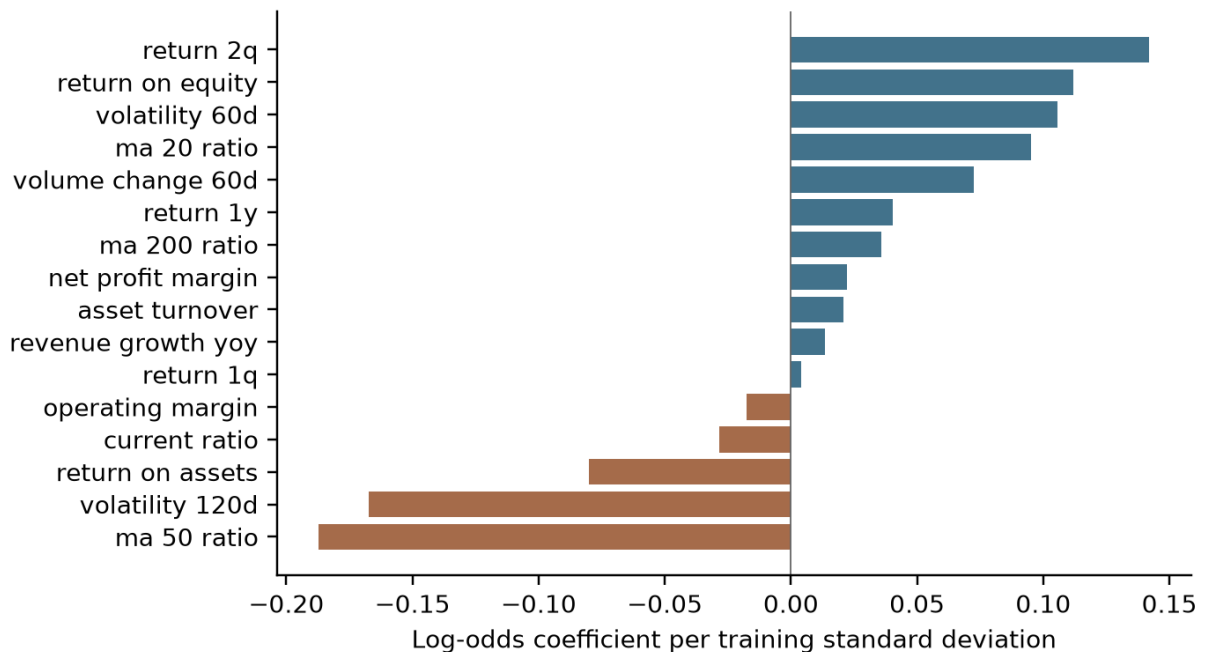


Figure 15. Logistic Regression coefficients after common clipping, imputation and standardisation. Coefficients describe the fitted model, not causal effects or significance tests.

Figure 15 provides an additive diagnostic alongside the nonlinear selected model. The 50-day price-to-mean deviation and 120-day volatility have the largest negative coefficients, approximately -0.187 and -0.167 . The 182-day return and return-on-equity proxy have positive coefficients of about 0.142 and 0.112 . These signs are conditional on the other included features and on the training preprocessing. The model's weak test AUC limits any substantive claim based solely on coefficient direction.

Impurity-based feature importance is also saved for Random Forest, Gradient Boosting and Decision Tree. These are fitted-model diagnostics; they are not external validation or statistical significance. Full numerical tables accompany the report so that the displayed summaries can be checked without relying on figure pixels.

5.3 Checks actually completed

Table 7. Completed computational checks and their limits.

Check	Observed result	Scope
Panel uniqueness	0 duplicate company-quarter keys	3,433 constructed records
Target endpoints	3,433 rows checked against cleaned daily series	Prices, dates and return-based labels
Availability timing	0 violations of implemented lag rules	Assumed lags; actual historical releases not certified
Label-boundary purge	58 rows removed; 0 residual boundary violations	Training and validation horizons
Future-price perturbation	Synthetic regression + 29 company checks pass after correction	Past-only market-feature wrapper
Preprocessing isolation	Training medians and 1%/99% cutoffs independently match	No test-derived preprocessing estimates
Model persistence	8 reloaded models reproduce saved test classes/probabilities	Same recorded environment and matrices
Metric verification	18 prediction tables match; pairwise AUC agrees	Eight models + baseline, validation and test
Convergence	No recorded training warnings	Does not establish a global optimum

Table 7 reports executed checks, not a promise of universal reproducibility. All models share the same 261 test keys and labels. The recorded environment is Python 3.12.13 with scikit-learn 1.7.2; resolved dependency versions and deterministic seed settings are retained in Appendix D and the supporting files. A second clean-machine training run was not performed. The historical source reports' claims of independent review are not used as evidence for the corrected model outputs.

The saved earlier predictions were separately recalculated and matched their archived metrics, but their feature code fails the future-price perturbation check. Those archived numerical results and the package's sensitivity experiments are not pooled with the corrected run. The source findings motivate the corrected method; they do not constitute an additional independent held-out sample.

6 Discussion

6.1 Accuracy, discrimination and class balance

The central empirical finding is a separation between two objectives. If the objective were only to maximise the number of correct test labels, the constant positive baseline is better than all eight fitted classifiers in this sample. If the objective gives equal importance to recognising positive and non-positive outcomes, SVM is the preferred fitted model under the validation rule and has the strongest test balanced accuracy. These statements are compatible because class prevalence influences ordinary accuracy but not the equal-class weighting in balanced accuracy.

SVM exchanges some true-positive detections for true-negative detections relative to the baseline. The baseline has zero false negatives but 110 false positives. SVM reduces false positives to 50 while introducing 67 false negatives. It gains 60 correctly identified non-positive quarters at the cost of 67 missed positive quarters. This explains the seven-observation accuracy deficit exactly. Whether that trade is desirable would depend on a specified decision problem and error costs, which this classification study does not estimate.

The positive-class F1 rankings are also affected by prevalence and decision bias. Random Forest and Gradient Boosting achieve higher F1 than SVM because they detect more positive outcomes. Their specificity is much poorer. The constant baseline has the highest F1 of all, yet never detects a non-positive quarter. A research conclusion based on F1 alone would therefore disregard an entire class of outcomes.

6.2 Why complexity does not establish superiority

The nonlinear models do not uniformly outperform the linear comparator, and the MLP's additional flexibility does not translate into superior held-out discrimination. This result is consistent with weak signal, data limitations or distribution shift, but the experiment does not identify which mechanism is dominant. Architecture comparisons are also confounded by differing class-weight support and fixed default complexity. The evidence is about these eight implemented estimators, not all possible members of their algorithm families.

The SVM's validation AUC of 0.578 falls to 0.520 in the test period. Its selected threshold still yields a balanced accuracy of 0.551, but the weaker ranking suggests caution about threshold portability. The result offers no basis for declaring a stable score-to-probability relationship or reliable confidence estimates for individual quarters. Calibration was not tested as a separate modelling objective.

6.3 Implications for this equity sample

The reconstructed features are broad numerical summaries. They omit many plausible drivers of pharmaceutical equity repricing and do not model a market benchmark, exchange rates or event timing directly. Pooling firms across countries provides observations but imposes a common prediction rule on different listings and reporting environments. The model comparison therefore describes this pooled feature representation rather than the full information content available to market participants.

Weak results are informative when the evaluation is explicit: they prevent inflated claims based on a favourable scalar score or a biased decision rule. They do not prove market efficiency, rule out every forecasting approach or establish that accounting data are irrelevant. A more defensible conclusion is that the current implemented

combination of trailing market variables and incomplete lagged fundamentals produces limited evidence of next-quarter directional discrimination.

7 Limitations

Historical information quality is the principal limitation. Financial statements are supplied as extracted histories rather than release-time vintages. The 61/121-day availability rules block use before the assumed date, but later revisions or restatements may remain in the source. Passing an availability inequality is therefore narrower than demonstrating complete freedom from look-ahead bias. This limitation cannot be removed by a train/test split alone.

The panel spans 1962–2026 because the later package uses all available market history. Different companies enter at different times, and several modern tickers may inherit predecessor histories. This creates unequal representation, long-run structural change and substantial missing fundamentals in early observations. No stationarity test or survivorship-bias correction establishes that these histories form a homogeneous population. The selected company universe is not a random sample of pharmaceutical firms.

Financial mapping is operational rather than exhaustive. It uses documented alias lists and normalises known unit scales, but does not reconstruct every line item from every annual-report page or parse Zenotech’s spreadsheet-only material. Annual/quarterly fallback changes the horizon of flow-to-stock proxies. Staleness limits prevent indefinite carry-forward but cannot supply missing accounting observations. Rovi’s absence from the implemented daily loader reduces the candidate universe from 30 to 29.

Adjusted prices and corporate-action conventions are not uniform economic objects across all source providers. Local calendar-date parsing, positive-price checks and deterministic source selection address concrete processing issues, but do not certify every historical adjustment. The target is not benchmark-adjusted, and pooling domestic-currency directions does not describe an international investor’s converted return. Thin trading may also affect the informational meaning of a recorded last price.

Model specifications are fixed from the later package. There is no exhaustive hyperparameter optimisation, repeated-seed comparison or nested cross-validation. The validation rule is explicit, but the archive had already exposed related empirical outcomes, making this a retrospective corrected reconstruction rather than a prospectively blinded study. The common held-out partition prevents direct fitting contamination; it does not erase prior knowledge of the project.

The test sample comprises only nine calendar quarters with 29 firms each. Outcomes within a quarter and within a company may be dependent. The report presents point estimates, not independence-based confidence intervals, statistical significance or claims of external replication. The saved sensitivity outputs use the earlier implementation and are deliberately excluded from this corrected comparison. No cost-aware backtest, turnover analysis or realised trading-profitability experiment is performed.

8 Conclusion

A complete computational comparison has been carried out for eight classifiers using a common rebuilt company-quarter panel, explicit target construction, training-only preprocessing and chronological evaluation. The final result contains actual row-level predictions and continuous scores, not placeholders or historical aggregate

substitutions. SVM is selected through validation balanced accuracy and AUC, and its persisted model reproduces the saved test outputs.

The empirical conclusion is restrained. SVM achieves 55.2% test accuracy and 55.1% balanced accuracy, with sensitivity of 55.6%, specificity of 54.5%, F1 of 0.589 and AUC of 0.520. No fitted model beats the 57.9% accuracy of the constant training-majority baseline. The preferred model is consequently a class-balanced research comparator with limited signal, not evidence of dependable investment performance.

The most valuable next analytical improvement would be a consistent, point-in-time fundamental dataset with documented releases, reporting frequencies and comparable coverage. A later study could pre-register a recent common sample, tune within chronological training folds and reserve a genuinely unseen future period. Such work should be evaluated as a new experiment. It is not represented as completed here, and it does not alter the present results.

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The supplied analytical package is the primary computational material: THESIS(4).zip, containing the pipeline, reorganised company data and implementation documentation. Its concrete estimator interfaces and scoring conventions are interpreted using the version 1.7 documentation (scikit-learn developers, 2025b). The supplied visual-reference reports inform presentation only and are not sources of this study’s empirical methods or findings.

Appendix A Company universe and coverage

The table lists the 29 represented companies. The training panel is unbalanced; every company contributes seven validation and nine test observations. Two boundary observations per company are retained in the panel as purged records and excluded from fitting/evaluation. Laboratorios Rovi is the candidate company without daily CSV input in the later package loader.

Table 8. Company-level coverage in the corrected reconstruction.

Company	Region	First quarter	Training	Validation	Test
Ajanta Pharma	India	2002-09	77	7	9
Alembic Pharmaceuticals	India	2011-09	41	7	9
Amgen	Non-Indian	1983-06	154	7	9
Aurobindo Pharma	India	1996-03	103	7	9
Baxter International	Non-Indian	1981-12	160	7	9
Biocon	India	2014-03	31	7	9
Biogen	Non-Indian	1991-09	121	7	9
Cipla	India	1996-03	103	7	9
Formycon	Non-Indian	2011-03	43	7	9
Fresenius SE	Non-Indian	1998-12	92	7	9
Glenmark Pharmaceuticals	India	2002-09	77	7	9
Jiangsu Hengrui Pharma	Non-Indian	2000-12	84	7	9
Jubilant Pharmova	India	2002-09	77	7	9
Lupin	India	2001-09	81	7	9
Marksans Pharma	India	2002-09	77	7	9
Merck & Co	Non-Indian	1962-03	239	7	9
Natco Pharma	India	1996-03	103	7	9
Novartis	Non-Indian	1996-12	100	7	9
Pfizer	Non-Indian	1972-06	198	7	9
Sanofi	Non-Indian	2014-03	31	7	9
Shanghai Fosun Pharma	Non-Indian	1998-09	93	7	9
Strides Pharma Science	India	2002-06	78	7	9
Sun Pharmaceutical	India	1996-03	103	7	9
Teva Pharmaceutical	Non-Indian	1982-03	159	7	9
Torrent Pharmaceuticals	India	2002-09	77	7	9
Viatis	Non-Indian	1980-03	167	7	9
Zenotech Laboratories	India	2004-03	71	7	9
Zhejiang Hisun Pharma	Non-Indian	2000-09	85	7	9
Zydus Lifesciences	India	2000-06	86	7	9

Appendix B Complete candidate feature dictionary

All windows below use calendar days. P is the last eligible price at or before the observation date; P-d is the last eligible price at or before d calendar days earlier. Current market calculations are restricted to past-only rows. Ratios are dimensionless. A zero denominator or unavailable input yields missingness, subsequently handled only by training-derived rules.

Table 9. Market candidate definitions and final status.

Feature	Formula	Status
return_1q	$P / P-91 - 1$	Retained
return_2q	$P / P-182 - 1$	Retained
return_1y	$P / P-365 - 1$	Retained
momentum_3m	Same definition as return_1q	identical to return_1q
momentum_6m	Same definition as return_2q	identical to return_2q
ma_20_ratio	$P / \text{mean}(\text{prices in prior 20 days}) - 1$	Retained
ma_50_ratio	$P / \text{mean}(\text{prices in prior 50 days}) - 1$	Retained
ma_200_ratio	$P / \text{mean}(\text{prices in prior 200 days}) - 1$	Retained
volatility_60d	Sample SD of daily returns within prior 60 days	Retained
volatility_120d	Sample SD of daily returns within prior 120 days	Retained
volume_change_60d	Mean volume in second half of observations in prior 60 days / mean volume in first half - 1	Retained

Table 10. Financial candidate definitions and final status.

Feature	Formula	Status
revenue_growth_yoy	$(\text{Revenue} - \text{same-frequency prior-year revenue}) / \text{prior-year revenue}; \pm 45\text{-day matching tolerance}$	Retained
net_profit_margin	Net income / revenue	Retained
operating_margin	Operating income / revenue	Retained
return_on_equity	Period net income / ending total equity; not annualised	Retained
return_on_assets	Period net income / ending total assets; not annualised	Retained
debt_to_equity	Total debt / total equity	all missing in training
current_ratio	Current assets / current liabilities	Retained
quick_ratio_proxy	Identical to current ratio; inventory is not subtracted	identical to current_ratio
asset_turnover	Period revenue / ending total assets; not annualised	Retained
price_to_book	Market capitalisation / total equity	all missing in training
price_to_earnings_proxy	Market capitalisation / net income	all missing in training

Financial source mapping uses the package's ordered aliases, including Total Revenues/Revenue/Net Sales variants for revenue, Net Income (Loss)/Net Income variants for net income, Operating Income/Operating Profit variants for operating income, Total Assets, recognised Total Equity variants, Total Current Assets and Total Current Liabilities. The full ordered alias list remains executable in

pipeline/_04_features_target.py within the extracted package; the corrected unit handling and row alignment are implemented in rebuild.py. No missing market-capitalisation or debt balance is invented.

Appendix C Resolved model parameters and numerical detail

The following compact records include every resolved estimator parameter exported before fitting. None is a post-test adjustment. Shared numerical preprocessing is described in Section 3; its complete fitted values are stored in preprocessing.json. Native class predictions and score orientation, not a retrospectively chosen threshold, determine each confusion matrix.

Logistic Regression

Table 11. Logistic Regression resolved settings.

Parameter values
C=1.0; class_weight=balanced; dual=False; fit_intercept=True
intercept_scaling=1; l1_ratio=None; max_iter=1000; multi_class=deprecated
n_jobs=None; penalty=l2; random_state=42; solver=lbfgs
tol=0.0001; verbose=0; warm_start=False

Random Forest

Table 12. Random Forest resolved settings.

Parameter values
bootstrap=True; ccp_alpha=0.0; class_weight=balanced; criterion=gini
max_depth=None; max_features=sqrt; max_leaf_nodes=None; max_samples=None
min_impurity_decrease=0.0; min_samples_leaf=1; min_samples_split=2; min_weight_fraction_leaf=0.0
monotonic_cst=None; n_estimators=100; n_jobs=None; oob_score=False
random_state=42; verbose=0; warm_start=False

Naive Bayes

Table 13. Naive Bayes resolved settings.

Parameter values
priors=None; var_smoothing=1e-09

Gradient Boosting

Table 14. Gradient Boosting resolved settings.

Parameter values
ccp_alpha=0.0; criterion=friedman_mse; init=None; learning_rate=0.1
loss=log_loss; max_depth=3; max_features=None; max_leaf_nodes=None
min_impurity_decrease=0.0; min_samples_leaf=1; min_samples_split=2; min_weight_fraction_leaf=0.0

Parameter values

n_estimators=100; n_iter_no_change=None; random_state=42; subsample=1.0
 tol=0.0001; validation_fraction=0.1; verbose=0; warm_start=False

SVM**Table 15. SVM resolved settings.****Parameter values**

C=1.0; break_ties=False; cache_size=200; class_weight=balanced
 coef0=0.0; decision_function_shape=ovr; degree=3; gamma=scale
 kernel=rbf; max_iter=-1; probability=True; random_state=42
 shrinking=True; tol=0.001; verbose=False

Neural Network / MLP**Table 16. Neural Network / MLP resolved settings.****Parameter values**

activation=relu; alpha=0.0001; batch_size=auto; beta_1=0.9
 beta_2=0.999; early_stopping=False; epsilon=1e-08; hidden_layer_sizes=[100]
 learning_rate=constant; learning_rate_init=0.001; max_fun=15000; max_iter=2000
 momentum=0.9; n_iter_no_change=10; nesterovs_momentum=True; power_t=0.5
 random_state=42; shuffle=True; solver=adam; tol=0.0001
 validation_fraction=0.1; verbose=False; warm_start=False

KNN**Table 17. KNN resolved settings.****Parameter values**

algorithm=auto; leaf_size=30; metric=minkowski; metric_params=None
 n_jobs=None; n_neighbors=5; p=2; weights=uniform

Decision Tree**Table 18. Decision Tree resolved settings.****Parameter values**

ccp_alpha=0.0; class_weight=balanced; criterion=gini; max_depth=None
 max_features=None; max_leaf_nodes=None; min_impurity_decrease=0.0; min_samples_leaf=1
 min_samples_split=2; min_weight_fraction_leaf=0.0; monotonic_cst=None; random_state=42
 splitter=best

Table 19. Exact confusion counts and positive forecast counts on the held-out test panel.

Model	TN	FP	FN	TP	Predicted up
Logistic Regression	54	56	76	75	131
Random Forest	35	75	47	104	179
Naive Bayes	86	24	126	25	49
Gradient Boosting	28	82	46	105	187
SVM	60	50	67	84	134
Neural Network / MLP	41	69	65	86	155
KNN	35	75	64	87	162
Decision Tree	39	71	63	88	159
Majority baseline	0	110	0	151	261

Appendix D Computational environment and reproduction

This run uses Python 3.12.13 on Windows 11. The resolved package versions below describe the actual executed reconstruction, not the archive author's presumed environment. The archive pins pandas 2.3.3 and NumPy 2.1.2; this reconstruction records its available runtime separately. Scikit-learn 1.7.2 is retained. Exact cross-version numerical equality with the archived run is not claimed.

Table 20. Executed numerical environment.

Package	Version
pandas	3.0.1
numpy	2.5.3
scikit-learn	1.7.2
scipy	1.18.1
pyarrow	23.0.1
matplotlib	3.11.1
joblib	1.6.0
threadpoolctl	3.6.0

Execution order: extract the supplied archive to reproduction/friend_package; install the versions in final_run/requirements.txt into an isolated environment; run rebuild.py; run verify_outputs.py and final_checks.py; run make_figures.py; then build_report.py and the document-rendering/QA scripts. The source root is derived from each script's location. Dependencies installed into reproduction/deps are used in this delivered workspace; a clean environment can instead install the recorded requirements normally.

rebuild.py rereads the source CSVs, constructs final_run/data/financial_long.csv, fundamentals.csv and panel.csv, then fits the estimators and saves their outputs. It never writes to the extracted input package. For a training-only repetition using the saved panel, the --train-only option is available. The primary saved model files include

the estimator and preprocessing state. This is a documented executable route; a second clean-machine run has not been completed.

Appendix E Supporting empirical files

Table 21. Supporting outputs for checking the reported results.

Location within reproduction/final_run	Contents
data/panel.csv	Company keys, dates, split, target endpoint prices/returns, availability fields and candidate features
predictions/*_validation.csv and *_test.csv	Eight-model row-level true labels, predictions, probabilities and oriented scores; baseline outputs also supplied
tables/metrics.csv	Complete validation/test metric comparison including TN, FP, FN and TP
tables/roc_*.csv	FPR, TPR and thresholds for every model and partition
models/*.joblib	Eight fitted estimators plus preprocessing state and feature order
tables/*importance.csv and logistic_coefficients.csv	Fitted feature diagnostics and validation permutation summaries
figures/*.png and *.pdf	Eight confusion/ROC panels and comparative/diagnostic figures
model_specs.json; preprocessing.json	Resolved estimator settings and fitted numerical transformations
leakage_checks.json; independent_verification.json; final_checks.json	Executed timing, endpoint, prediction and metric checks
requirements.txt; environment.json	Recorded numerical environment

The full prediction files are the numerical appendix for individual observations; they are not replaced by a few examples in the printed report. Each figure in the empirical chapter is derived from these files. All draft backups and original source materials are preserved separately.